SET B

11	Perform the following: a. $(1101010)_2 \rightarrow (\ )_{10}$ b. $(10110101)_2 \rightarrow (\ )_{16}$ c. $(1011010)_2 + (10101)_2$	4	3	1	2	2.4.1
12	Show how to implement the AND, OR, and NOT gates using only NAND gates. Provide detailed diagrams and explanations for each implementation.	4	2	1	2	2.4.1
13	Convert the following from BCD to Gray Codes: a. $(1101)_2$ b. $(10010)_2$	4	3	1	2	2.4.1
14	Illustrate the process involved in ALU with a neat sketch.	4	2	2	2	2.4.1
15	Write the working principle of RAM in memory management and how it is important in terms of speed and process.	4	2	2	2	2.4.1

Part – C
(2 x 12 = 24Marks)

16	a) Explain how different number systems (binary, decimal, hexadecimal) can be used to encode and transmit data securely. Explain the steps of how the data is converted. b) Discuss the binary codes (such as BCD, Excess-3, and Gray) representation in data transmission.	12	4	1	2	2.4.1

Decimal	67	182
Binary	?	?
Octal	?	?
Hexadecimal	?	?

Decimal	26	43
BCD	?	?
XS-3	?	?
Gray	?	?

OR

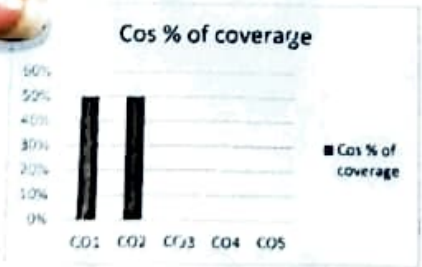
17	Perform the following operations and explain each step in detail: a. Add the binary numbers 110101 and 1011 using 1's complement representation. b. Subtract the binary number 1101 from 1010 using 2's complement representation. c. Multiply the binary numbers 101 and 110 using the standard binary multiplication method. d. Divide the binary number 110100 by 101 using binary division.	12	4	1	2	2.4.1
18	Consider a computer system with a different bus structure that includes the following components: 1. Data Bus 2. Address Bus 3. Control Bus Describe the role of each component in the bus structure and explain how they interact to facilitate communication between the CPU, memory, and I/O devices.	12	2	2	2	2.4.1

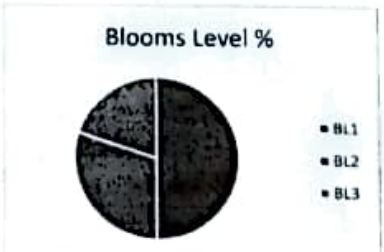
OR

19	Explain different types of instructions used in computer organization. Evaluate (A+B) * (C+D) using zero, one, two and three address instructions.	12	2	2	2	2.4.1

***Performance Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.**

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator